Nikhil Maan

+91-9310028159 Email Linkedin Github Portfolio

Education

Netaji Subhas University of Technology **2023 – 2027**
Bachelor of Technology in Computer Science and Artificial Intelligence *New Delhi, India*

Dunes International School **2022 – 2023**
12th grade, 90 *Abu Dhabi, UAE*

ADIS, Al Wathba **2020 – 2021**
12th grade, 90.4 *Abu Dhabi, UAE*

Technical Skills

Languages: Python, C/C++, Java, JavaScript, Go, Bash, SQL
AI/ML & Quantum: PyTorch, PennyLane (QML), XGBoost, Scikit-learn, OpenCV, CNNs, ViTs
Developer Tools: Git, Docker, Linux (Bash), LaTeX, High-Performance Computing (HPC)
Web/Frameworks: Three.js, React, Node.js, Spring Boot, MySQL, MongoDB, Firebase

Publications

QVIT-RQC: Hybrid Quantum-Classical ViT for Brain Tumor Diagnosis **Under Review**

- **N. Maan**, et al. "QVIT-RQC: A Hybrid Quantum-Classical Vision Transformer with Modulated Attention and Residual Classification for Brain Tumor Diagnosis." Submitted to **Springer Nature**, 2026.
- Developed a 4-qubit VQC for dynamic attention (HQA) and a residual classifier (RQC) to achieve a SOTA **99.06% accuracy** on 3,064 MRI images.

Multi-modal Alzheimer's Diagnosis using CNN and XGBoost **Accepted**

- **N. Maan**, et al. "A Multi-modal approach for Alzheimer's Diagnosis using CNN and XGBoost from MRI Imaging and Tabular Biomarkers." **Accepted at NIT Delhi Conference**, 2026.
- Achieved **98.8% accuracy** on MRI data using Inception-based CNNs and **98.58%** on clinical data via XGBoost.

Projects

Relevant LeetCode Solutions | *Algorithms, Data Structures* **Live**

- Maintained a curated collection of algorithmic solutions, focusing on data structures and problem-solving.
- Gained 200+ GitHub stars from contributors and learners worldwide.
- Independently maintained and updated repository for consistent practice.
- Repository: github.com/nikhilm25/RelevantLeetcode

Eco Twin: Urban Air Management | *React, Node.js, Three.js, MongoDB* **Github**

- Selected as one of the top 5 teams nationally in the SIH Grand Finale for developing a Digital Twin platform using Three.js to simulate industrial hardware with real-time physics feedback.
- Engineered a geospatial dashboard with Leaflet and Heatmap overlays to visualize emission hotspots and perform source attribution for point and mobile pollutants.
- Integrated an AI-Driven Decision Support System utilizing live sensor telemetry and Open-Meteo APIs to recommend optimal carbon mitigation strategies via cost-benefit analysis.
- Built a secure backend with JWT-based RBAC and OTP verification, managing complex NoSQL schemas for multi-city project tracking and historical trend analysis.

Shell on the JVM | *Java* **Github**

- Implemented a Unix-like shell in Java with command parsing and execution engine.
- Included support for built-in commands as well as history and logging.
- Added support for process management and I/O redirection.
- Ensured cross-platform compatibility through the JVM.

Achievements

- Solved 400+ LeetCode problems, ranking in the Top 11% globally in contests.
- Secured 2nd place among 1000+ participants in Code Showdown, showcasing exceptional algorithmic abilities
- Selected as a National Grand Finalist in Smart India Hackathon (SIH) 2024 for 'Eco Twin', ranking among the top 5 teams nationally for urban air quality innovation.